

1. Smart Inventory Management System

Problem Statement

An e-commerce warehouse stores product information in an SQLite database.

Each product has:

- * Product ID
- * Product Name
- * Category
- * Quantity
- * Price

Requirements

1. Fetch all products from the SQLite database.
2. Store them in Python objects.
3. Sort the products based on quantity using ****Merge Sort****.
4. Allow the manager to search for a Product ID using ****Binary Search****.
5. Display the complete product details.
6. Display products whose stock is below 10.

Concepts

- * SQLite
- * Python Classes
- * Merge Sort
- * Binary Search
- * SQL SELECT

2. Hospital Patient Queue Management

Problem Statement

A hospital stores patient details in SQLite.

Each patient has:

- * Patient ID
- * Name
- * Age
- * Priority Level

Requirements

1. Fetch all patients.
2. Create a ****Priority Queue**** based on Priority Level.
3. Attend patients in order of priority.
4. After attending a patient, update the database.
5. Display the remaining patients.

Concepts

- * SQLite
- * Priority Queue
- * UPDATE Query
- * Heap/Priority Queue

3. Online Banking Transaction Analyzer

Problem Statement

A bank stores transactions in SQLite.

Each transaction contains:

- * Transaction ID
- * Account Number
- * Amount
- * Date
- * Type (Credit/Debit)

Requirements

1. Retrieve all transactions.
2. Sort them by amount using **Quick Sort**.
3. Search transactions using Transaction ID.
4. Calculate total credits and debits.
5. Display the top 5 highest-value transactions.

Concepts

- * SQL
- * Quick Sort
- * Binary Search
- * Aggregation

4. College Placement Portal

Problem Statement

A college stores student records in SQLite.

Each student contains:

- * Roll Number
- * Name
- * CGPA
- * Skills
- * Placement Status

Requirements

1. Retrieve all students.
2. Sort students by CGPA using **Heap Sort**.
3. Search students by Roll Number.
4. Display students eligible for placements (CGPA > 7.5).
5. Update placement status after selection.

Concepts

- * Heap
- * Heap Sort
- * Binary Search
- * SQL UPDATE

5. Movie Recommendation System

Problem Statement

A streaming platform stores movie information.

Each movie contains

- * Movie ID
- * Title
- * Genre
- * Rating
- * Watch Count

Requirements

1. Fetch all movies.
2. Sort movies based on Rating.
3. Search a movie using Movie ID.
4. Display Top 10 highest-rated movies.
5. Display the most watched movie in every genre.

Concepts

- * Sorting
- * Searching
- * Dictionaries
- * SQL GROUP BY

6. Food Delivery Route Optimizer

Problem Statement

A food delivery company stores restaurants and delivery locations.

Database tables:

Restaurant

Delivery

Orders

Requirements

1. Fetch all pending orders.
2. Build a **Graph** representing restaurants and delivery areas.
3. Find the shortest delivery path using **BFS**.
4. Display delivery order sequence.
5. Mark completed deliveries in SQLite.

Concepts

- * Graph
- * BFS
- * SQL JOIN
- * UPDATE

7. Employee Attendance Analytics

Problem Statement

A company records employee attendance.

Each record contains:

- * Employee ID
- * Name
- * Check-in Time
- * Check-out Time

Requirements

1. Retrieve attendance records.
2. Calculate total working hours.
3. Sort employees by total hours worked.
4. Search employees using Employee ID.
5. Display employees who worked more than 45 hours this week.

Concepts

- * SQL
- * Sorting
- * Binary Search
- * DateTime

8. Ride Booking Management System

Problem Statement

A cab company stores:

Drivers

Customers

Bookings

Requirements

1. Fetch available drivers.
2. Store them in a Graph representing city connectivity.
3. Find the nearest available driver using **BFS**.
4. Assign the booking.
5. Update driver availability.

Concepts

- * Graph
- * BFS
- * SQL JOIN
- * UPDATE

9. Library Book Borrowing System

Problem Statement

A library stores books and borrowing history.

Tables:

Books

Members

Borrowed Books

Requirements

1. Display all available books.
2. Sort books alphabetically using ****Merge Sort****.
3. Search books by Book ID.
4. Borrow a book.
5. Update book availability.
6. Display overdue books.

Concepts

- * Merge Sort
- * Binary Search
- * SQLite
- * SQL UPDATE

10. Sales Performance Dashboard

Problem Statement

A retail company stores sales information.

Each sale contains:

- * Salesperson ID
- * Product
- * Quantity
- * Revenue
- * Region

Requirements

1. Retrieve all sales records.
2. Sort records based on revenue.
3. Search a salesperson using Employee ID.
4. Display Top 5 salespersons.
5. Find the highest revenue region.
6. Update monthly incentive status for eligible salespersons.

Concepts

- * SQL GROUP BY
- * Sorting
- * Binary Search
- * Dictionaries
- * SQLite